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# Multistate reweighting and configuration mapping together accelerate the efficiency of thermodynamic calculations as a function of molecular geometry by orders of magnitude

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We present an approach to calculate free energy and other thermodynamic property differences between molecules which have very little or no overlap in configuration space, but where a one-to-one mapping between the molecule geometries exists. The approach combines multistate reweighting with remapping of phase space between simulated states. We apply this method to calculate the free energy differences between non-overlapping, truncated harmonic oscillators, the free energy, enthalpy, and entropy differences between different parameterizations of rigid water, and differences in free energy of solvation between dipoles of different lengths. Previously difficult or impossible problems become either trivially easy or are improved in efficiency by two to five orders of magnitude. © 2013 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4801332>]

## I. INTRODUCTION

Computer-aided molecular design problems like drug design,<sup>1</sup> solvent design,<sup>2</sup> and molecular recognition<sup>3</sup> require high throughput screening of a large library of chemical compounds. Thermodynamic properties of interest such as solvation free energy or binding free energy thus must be calculated over a large multidimensional chemical space which may involve significant geometrical changes within molecules. Similarly, refining force field parameters based on solvation free energies requires high throughput thermodynamic calculations over a large multidimensional parameter space made of both interaction and geometry parameters.

For a given accuracy of atomistic force fields, there are major challenges that stand in the way of free energy calculations from becoming a mainstream tool for *in silico* molecular design. The current high computational costs associated with generating samples from simulation make free energy calculations impractical for screening large numbers of compounds using molecular dynamics or Monte Carlo simulations. This is especially true for structurally different compounds, for which methods to calculate free energy differences exist but are not straightforward to implement in a high throughput manner.

In this study, we show that for problems involving changes of molecular geometry, we can reduce the computational sample generation costs of calculating free energies and other properties by up to five orders of magnitude by adapting the reweighting formalism of multistate Bennett acceptance ratio (MBAR)<sup>4</sup> to include phase space mapping. MBAR can predict thermodynamic properties for poorly sampled or even unsampled states using data from only a few sampled states. However, the present form of MBAR cannot be used to estimate free energy and expectation value differences between geometrically different molecules which have little or no overlap in configuration space because of these geometrical differences. In this paper, we demonstrate how to extend

multistate reweighting such as MBAR to perform free energy calculations for molecules of different geometries and demonstrate the significant computational efficiencies that can be achieved.

Methods to compute free energies and expectations based on potential reweighting, such as exponential averaging,<sup>5</sup> the Bennett acceptance ratio method,<sup>6</sup> and MBAR<sup>4</sup> require that all samples from the ensemble of configurations generated from one thermodynamic state should be observed with nonzero probability in the configurational ensemble of the other thermodynamic states of interest for the calculations to converge to the correct answer. When there is poor overlap, many samples are required to eliminate bias from the calculation and the calculations will have large statistical uncertainty. When there is zero overlap in configuration space, no amount of sampling results will give an unbiased free energy difference. For example, if we wish to calculate the free energy difference between a SPC/E<sup>7</sup> and a TIP3P<sup>8</sup> water molecule, we will never see a rigid TIP3P molecular configuration in rigid SPC/E simulation. Thus, reweighting methods would fail and we will get incorrect free energy differences and in most cases, large uncertainties as well.

One approach to overcome this problem of non-overlapping phase spaces is to perform sampling with a potential that explicitly includes phase space overlap with both ends of the free energy calculation. For example, Oostenbrink and van Gunsteren developed a single step perturbation method allowing efficient calculation of free energy differences between sufficiently similar molecules.<sup>9</sup> This method involves choosing a reference structure,  $R$ , such that simulations of  $R$  share phase space with both end state structures  $A$  and  $B$ . The free energy difference  $\Delta G_{AB}$  can be calculated as difference between  $\Delta G_{AR}$  and  $\Delta G_{BR}$ . However, the selection and design of reference states for large number of compounds or reaction coordinates is a complicated problem with no clear comprehensive solution. More general investigations of designing

intermediates with phase space overlap have been explored by Kofke and co-workers.<sup>10–13</sup>

Straatsma, Zacharias, and McCammon<sup>14</sup> proposed a potential of mean force approach using thermodynamic integration (TI) and includes holonomic constraints. They calculated the free energy along a chosen reaction coordinate using the constraint forces evaluated from the SHAKE<sup>15</sup> coordinate correction. The method has been successfully used in a number of free energy calculation applications.<sup>16–20</sup> Such TI-based free energy calculations make use of analytical constraint derivatives with respect to one or two reaction coordinates.<sup>21</sup>

Other methods to calculate free energy differences for geometrically diverse compounds, both rigid and nonrigid, have been proposed based on constructing phase space mappings between two end states.<sup>14,22</sup> With this approach, a coordinate transformation is defined between each point in the phase space of each molecular system. Since all free energies and expectation values are calculated by integrals over the respective phase space, the coordinate transformations result in Jacobian terms that can be integrated into the free energy difference calculations.

Several other types of mathematical transformations between the phase spaces of two molecular states have been proposed in order to improve the efficiency of calculations for free energies and other ensemble observables. For example, Voter<sup>23</sup> introduced a translational shift in the integration variable to map between two energy distributions in order to estimate free energies for states separated in configurational space. Voter's method, however, only works for spatially translated distributions. Ytreberg and Zuckerman generalized Voter's method to include translations of internal coordinates as well as incorporate them idea into the Bennett acceptance ratio.<sup>24</sup>

To deal with more complex and general transformations, Jarzynski<sup>22</sup> proposed the targeted free energy perturbation approach, using an invertible transformation of the phase space  $A$  which maps to a new phase space  $A'$  such that  $A'$  has significant overlap with the phase space of the target distribution  $B$ , and later extended this to time-dependent mappings.<sup>25,26</sup> Jarzynski applied this method to calculate the free energy associated with the expansion of cavity in a fluid. He envisioned his technique could be used for free energy estimation over different molecular geometries. However, a way to use this formalism for molecular problems was not explicitly laid out. The general approach described by Jarzynski is similar to that of Severance, Essex, and Jorgensen,<sup>27</sup> where a Jacobian factor was included to account for phase space contributions to the free energy of changing bond lengths, but only for unidirectional perturbation formulas.

Tan, Schultz, and Kofke applied this formalism to a realistic molecular problem,<sup>28</sup> studying the temperature dependence of solid phase free energies by combining Jarzynski's targeted free energy method with the Bennett acceptance ratio.<sup>6</sup> Because the harmonic portion of the potential is dominant, configurations observed at different temperatures could therefore be mapped to each other with respect to deviation from a temperature dependent harmonic reference potential, and the much smaller differences from this

temperature-dependent harmonic reference potential could be calculated with lower uncertainty.

Similar configuration mapping methods have been investigated by researchers in statistics as well. MBAR itself can be seen as application of extended bridge sampling.<sup>4,29</sup> Meng and Shilling<sup>30</sup> developed a set of methods similar to Jarzynski's mapping technique to estimate the ratio of normalizing constants, which they called "warp bridge sampling." They proposed three types of warp-bridge mapping. Warp-I maps distributions by shifting the integration variable. Voter's method therefore belongs to Warp-I category. Warp-II involves mapping the variance as well as the mean, which is equivalent to Tan *et al.*'s use of harmonically targeted temperature perturbation to achieve phase space overlap.<sup>28</sup> Warp-III involves mapping mean, spread, and skewness of the distributions.

We propose a general algorithm inspired by these previous mapping approaches, specifically combining mapping of configurations via a change of variables with the MBAR, the minimum variance multistate reweighting method. MBAR is provably the most efficient method to estimate free energies and expectation values<sup>4</sup> (although if carefully handled, the drawbacks of other standard methods can be minimized), and has been demonstrated to have lower bias than thermodynamic integration in the limit of fewer alchemical intermediate states.<sup>31</sup>

Using MBAR with configuration mapping allows the use of data from every sampled state to estimate free energy difference between any two given states, even when these states differ in the geometry. The configurations at any sampled state are mapped to geometries of the other states, creating phase space overlap in the configurational space. A Jacobian factor due to the change of variables is required from the mapping, but can usually be computed in a postprocessing step.

The advantage of MBAR with mapping is that it is easy to handle multiple geometries once pairwise maps implementing the invertible transformations are known. Free energy differences and other physical properties can then be efficiently estimated for a large number of thermodynamic states having different geometries without modifying anything in the simulation code used for sample generation by postprocessing the simulation data.

We demonstrate this approach to free energy calculations by applying it to a toy model (truncated harmonic oscillators) as well to realistic rigid water models, specifically transformations between SPC/E,<sup>7</sup> TIP3P,<sup>8</sup> and TIP4P,<sup>8</sup> and to the change in equilibrium bond length of a dipole formed by partial charges separated by a harmonic bond.

In the harmonic oscillator example, we vary two parameters, the force constant  $k$  and the equilibrium position  $\mu$ , between different states. Because these harmonic oscillators are truncated, there is exactly zero phase space overlap between any two systems. However, despite this lack of overlap, free energy differences and expectation values can be calculated to exact agreement with analytical results with any number of samples by explicitly constructing a mapping between the oscillators.

Similarly, rigid water models sample from completely different configuration spaces, as we will never see a TIP3P

geometry while performing a simulation of TIP4P. However, we can construct a one-to-one mapping between configurations of any three- or four-point water model which preserves the center of mass of the molecule. Indeed, such a mapping can be constructed between any rigid water model sharing the same symmetry, regardless of how many interaction sites are included. Although the phase spaces sampled by the different water molecules do not have sufficiently large overlap to perform this transformation in a single step, we can construct a series of intermediate states that have reversible maps to each of the end states and each other, linearly interpolating all other parameters, and then compute free energies and thermodynamic property differences using data from all the intermediates simultaneously. We examine an increasing number of intermediates, and find we can calculate unbiased results with very high precision with as few as one to three intermediate states.

Finally, we calculate the free energy difference between two dipoles of equal charge but different equilibrium bond length and show that even for flexible molecules these mapping calculations can drastically increase the efficiency of such transformation calculations.

In all cases, we can construct thermodynamic cycles to check the results that show complete cancellation to within the calculated errors. Overall, we find that we can drastically improve the estimates of differences in properties between the molecular models. This improvement is so great that it becomes practically impossible to validate the errors by running standard free energy simulations because of the  $N_{smp}^{-1/2}$  dependence of the statistical error on the number of samples  $N_{smp}$  collected. The alternate approach we present is simply that much more efficient.

## II. THEORY

We define a reduced energy  $u_i(q) = \beta_i(H_i(q, p) + P_i V - \sum_k \mu_{k,i} N_k)$ , corresponding to state  $i$  with Hamiltonian  $H_i$ , pressure  $P_i$ , chemical potentials of  $k$  species  $\mu_{k,i}$ , multidimensional coordinates and velocities  $q$  and  $p$ , volume  $V$ , and species numbers  $N_k$ . The thermodynamics state  $i$  is defined by the allowed microstate configurations  $x = \{p, q, V, N_k\}$  defined over the phase space volume  $\Gamma_i$  with microstate probabilities determined by the Boltzmann probability distribution  $P_i(x) \propto \exp(-u_i(x))$ .

We define a mapping  $T_{ij}$  which takes a configuration  $x_i$  sampled from the phase space  $\Gamma_i$  of thermodynamic state  $i$  to  $T_{ij}(x_i)$ , a sample contained in the phase space  $\Gamma_j$  of thermodynamic state  $j$ . To emphasize, the convention we use is that  $T_{ij}$  carries from initial phase space  $i$  to final phase space  $j$ . We assume that this map is completely invertible, such that  $T_{ij}^{-1} = T_{ji}$  exists, and that it is bijective, i.e., it is one-to-one and onto between phase space  $\Gamma_i$  and  $\Gamma_j$ .  $T_{ij}$  is allowed to depend on  $x_i$  (and likewise  $T_{ji}$  on  $x_j$ ) though our examples will use linear maps that do not depend on the coordinates  $x_i$  or  $x_j$ . The transformation  $T_{ij}$  will have an associated Jacobian matrix  $J_{ij}(x_i)_{kl} = \frac{\partial(T_{ij}(x_i))_k}{\partial x_{il}}$ , where  $(T_{ij}(x_i))_k$  is the  $k$ th component of the  $x_j$  coordinate space, and  $x_{il}$  is the  $l$ th component of the  $x_i$  coordinate space.

This matrix may be a function of  $x_i$ , the coordinates of the domain, and because the transformation is invertible, we will have the relationship  $J_{ij}(x_i) = J_{ji}^{-1}(x_j)$ .

Our central finding is that if we define a “warped reduced energy” term which includes the Jacobian term:

$$u_{ij}^w(x_j) = u_i(T_{ji}(x_j)) - \ln |J_{ji}(x_j)| \quad (1)$$

all the multistate reweighting identities derived in previous papers in the context of the multistate Bennett acceptance ratio method will hold. We can then re-write the multistate reweighting equation in terms of warped reduced energy terms,

$$f_i = -\ln \left( \sum_{j=1}^K \sum_{n=1}^{N_j} \frac{\exp(-u_{ij}^w(x_{jn}))}{\sum_{k=1}^K N_k \exp(f_k - u_{kj}^w(x_{jn}))} \right), \quad (2)$$

regaining the same form as the original multistate Bennett acceptance ratio. Once we define the mappings and have the  $K \times K \times N_k$  warped energies  $u_{ij}^w(x_j)$  we can then solve Eq. (2) self-consistently to find all free energies using methods already introduced.<sup>4</sup> Expectation values of arbitrary functions can be computed using the same formalism. Indeed, all multistate reweighting formulas developed by Shirts and Chodera remain valid,<sup>4</sup> with the proof presented in the Appendix. This result is not totally unexpected; as we have discussed, similar single state reweighting formulas written in terms of a “warped” potential as described in Eq. (1) have been developed for analyzing molecular simulations, and a similar multistate formulation has been observed in statistics, as discussed in the Introduction. However, this is the first full generalization of this idea to calculations involving multiple thermodynamics states, and we demonstrate for the first time that the combined multistate plus configuration space mapping approach can make previous challenging problems of computing differences in observables as a function of molecular transformation much easier.

## III. MODELS

### A. Truncated harmonic oscillators

We first examine the problem of calculating the free energy difference between a system of truncated (or censored) harmonic oscillators with spring constants  $k_i$ , truncated at one standard deviation  $\sigma_i = k_i^{-1/2}$  from their mean value  $\mu_i$ , with the potential becoming infinite (and probability zero) with any larger displacement, so that

$$U_i(x) = \begin{cases} \frac{1}{2}k_i(x - \mu_i)^2 & |x - \mu_i| < \sigma_i \\ \infty & |x - \mu_i| \geq \sigma_i \end{cases}$$

We assume that the spring constants  $k_i$  are chosen sufficiently large and harmonic oscillator potential centers  $\mu_i$  are chosen sufficiently far apart that there is no phase space overlap between any two oscillators. In this case, no amount of sampling will allow the calculation of the free energy differences between these oscillators by any reweighting method.

We now approach this problem with the assistance of the mapping formalism. We first must define our maps  $T_{ij}$ . A general approach to construct pairwise maps is to construct each map  $T_{ij}$  implicitly by constructing maps  $T_i$  to a reference state, and composing maps to obtain maps  $T_{ij}(x_i) = T_j^{-1}T_i(x_i)$ . Although this problem is simple enough not to require this approach, we will use it for pedagogical purposes, as it becomes useful for more complicated problems. For this toy problem, we use a reference state with zero mean and unit variance for the underlying Gaussian that is then truncated, and we map by shifting and rescaling:

$$T_i(x_i) = \frac{x_i - \mu_i}{\sigma_i}, \quad (3)$$

$$T_{ij}(x_i) = T_j^{-1}(T_i(x_i)) = \left( \frac{x_i - \mu_i}{\sigma_i} \right) \sigma_j + \mu_j.$$

This transformation  $T_{ij}(x_i)$  translates the sample by  $-\mu_i$ , rescales by  $\sigma_j/\sigma_i$  and then translates by  $+\mu_j$ . The Jacobians  $J_{ij}$  and  $J_{ji}$  for this map are

$$J_{ij} = \frac{\partial T_{ij}(x_i)}{\partial x_i} = \frac{\sigma_j}{\sigma_i}, \quad (4)$$

$$J_{ji} = \frac{\partial T_{ji}(x_j)}{\partial x_j} = \frac{\sigma_i}{\sigma_j}.$$

In this case the maps and the corresponding Jacobians are independent of the position coordinate.

Calculation of  $u_i(x_{jn})$  is not possible without the mapping since  $x_{jn}$  is never seen in the  $i$ th simulation. The warped reduced energy  $u_{ij}^w(x_{jn})$  using the mapping and Jacobian terms is

$$\begin{aligned} u_{ij}(x_{jn}) &= \beta \frac{1}{2} K_i (T_{ji}(x_{jn}) - \mu_i)^2 - \ln \frac{\sigma_i}{\sigma_j} \\ &= \beta \frac{1}{2} K_i \left( \frac{\sigma_i(x_{jn} - \mu_j)}{\sigma_j} + \mu_i - \mu_i \right)^2 - \ln \frac{\sigma_i}{\sigma_j} \\ &= \beta \frac{1}{2} K_i \left( \sqrt{\frac{K_j}{K_i}} (x_{jn} - \mu_j) \right)^2 - \ln \frac{\sigma_i}{\sigma_j} \\ &= u_j(x_{jn}) - \ln \frac{\sigma_i}{\sigma_j}. \end{aligned} \quad (5)$$

These energies can then be used in Eq. (2), including Jacobian term, and free energies estimated and validated against the analytical estimates of free energies. The analytical estimate of the Helmholtz free energy of a truncated harmonic oscillator at constant temperature depends only on the force constant with  $A_{i,\text{analytical}} = -k_B T \ln(\sqrt{2\pi}\sigma_i \text{erf}(2^{-1/2}))$ .

In this particular case, the free energy differences between two harmonic oscillators are simply  $A_i/A_j = -k_B T \ln \sigma_i/\sigma_j$ . We note that the potential energy will not change after mapping from  $x_j$  to  $T_{ji}(x_j)$ , so the potential energy contribution to the free energy from all sampled points will be zero, leaving simply the Jacobian contribution. We, therefore, require exactly zero samples in order to correctly predict the free energy, since we have an exact phase space mapping which takes each sample to a sample having equal Boltzmann weight as before the mapping. This exact phase space mapping for our truncated harmonic oscillators is an example of the ‘‘Warp-II’’

strategy of Meng and Schilling,<sup>30</sup> with a transformation scaling both the coordinates and the variance of the distribution of energies.

Multistate reweighting using MBAR also makes it possible to evaluate not only free energies but also expectations at state  $i$ , using the sampling performed at all of the  $K$  states, rather than calculating expectation values at state  $i$  using only samples taken from state  $i$ . When using multistate reweighting with mapping for this truncated harmonic oscillator problem, the statistical uncertainty of any given observable (such as average potential energy or RMS fluctuation around the mean position) scales as  $(\sum_i N_i)^{-1/2}$ , the total number of samples collected among all states, rather than the number of samples  $N_i$  collected at each state, because each sample, from any state, contributes equal information to ensemble averages at all other states. This is because the phase space mapping is perfect; each point in phase space, after mapping, has the same energy, and thus same Boltzmann weight. If the mapping is not perfect then, the error in the expectation values will have some intermediate efficiency, scaling better than  $N^{-1/2}$  but not as good as  $(\sum_i N_i)^{-1/2}$ .

In almost all cases of real physical relevance, such an exact mapping cannot be constructed. Although one-to-one mappings between spaces can be easily constructed in many cases (as we will show), they very rarely map between points with equal Boltzmann weight. However, this mapping technique can be used to create overlap where this overlap did not exist before. For example, if the harmonic oscillators were truncated at an arbitrary distance  $C_k$  for all oscillators rather than at 1 standard deviation for all states, then a new map  $T_{ij}(x) = (\frac{x_j - \mu_j}{C_j})C_j + \mu_i$  with Jacobian  $C_i/C_j$  can be used. The potential energies will change under such a transformation, but there will be a moderate amount of phase space overlap as long as the force constants are not very different, allowing MBAR to be used to calculate free energies and expectation values efficiently.

Alternatively, mappings can be used to increase overlap where less overlap initially existed. For example, if we had standard harmonic oscillators, without any truncation, then there will always exist some nonzero overlap between any two probability distributions, as the normal distribution is nonzero (if small) everywhere. However, because of the poor phase space overlap if the harmonic oscillators have high spring constants, or are separated by a large distance, it can take a very large number of samples to calculate free energy differences with a moderate degree of precision. In this case, if we happen to use the same  $T_{ij}(x)$  defined in the case of truncated harmonic oscillators, we will again have an exact mapping, meaning free energy differences require no collection of samples, and samples collected at any state will be equally good for calculating expectation values for all states.

## B. Water models

We next demonstrate how multistate reweighting with mapping can be used to drastically improve the efficiency of molecular transformation calculations. We calculate the pairwise difference in the Gibbs free energy of hydration  $\Delta\Delta G_{\text{hyd}}$

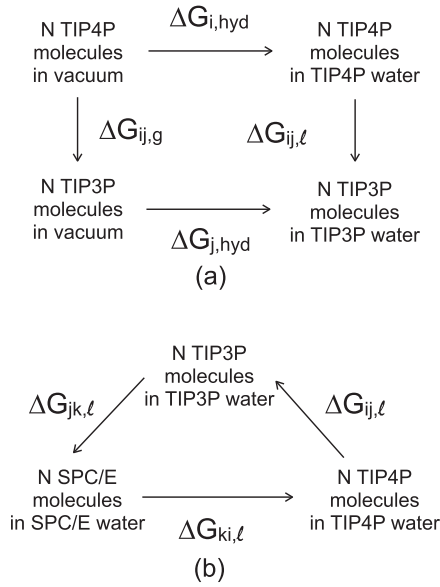


FIG. 1. The difference between analytically calculated free energy  $\Delta G_{ij,g}$  and free energy calculated using MBAR with mapping  $\Delta G_{ij,\ell}$  should be consistent with the difference in the solvation free energies calculated using MBAR without mapping for thermodynamic cycle (a) according to Eq. (6) for the three transformations.  $\Delta G_{ij,\ell}$ ,  $\Delta G_{jk,\ell}$ , and  $\Delta G_{ki,\ell}$  should add up to zero in thermodynamic cycle and (b) according to Eq. (7).

as well as differences in enthalpy and entropy between three water models using both standard multistate reweighting and multistate reweighting with mapping. The thermodynamic cycle shown in Figure 1 relates these quantities. For concreteness, it is shown for TIP4P to TIP3P, but same cycle can be constructed for any pair of rigid water models (such as TIP3P to SPC/E or SPC/E to TIP4P).

In this thermodynamic cycle  $\Delta G_{ij,g}$  is the Gibbs free energy of mutating  $N$  water molecules of model  $i$  to  $N$  molecules of model  $j$  in the ideal gas state. Interaction between molecules in the ideal gas state is zero. Thus the reduced warped potential in Eq. (1) will simply be equal to the PV term minus the Jacobian term  $|J_{ji}|$ .

$\Delta G_{i,hyd}$  in the thermodynamic cycle (a) is the free energy of hydration of  $N$  molecules of model  $i$ . The solvation free energy of  $N$  molecules is  $N$  times the chemical potential  $\mu_{i,\ell}$  (the solvation free energy for a single molecule of model  $i$  in pure liquid), using the ideal gas reference state.  $\Delta G_{ij,\ell}$  is the free energy of mutating  $N$  water molecules of model  $i$  to  $N$

molecules of model  $j$  in the liquid state.  $\Delta G_{ij,\ell}$  is calculated using the MBAR with mapping formalism.  $\Delta G_{ij,\ell}$ ,  $\Delta G_{i,hyd}$ ,  $\Delta G_{j,hyd}$ , and  $\Delta G_{ij,g}$  follow a thermodynamic cycle given by Eq. (6),

$$\Delta G_{i,hyd} - \Delta G_{j,hyd} = \Delta G_{ij,\ell} - \Delta G_{ij,g}. \quad (6)$$

Dividing the entire Eq. (6) by  $N$  gives the difference in free energy changes for a single molecule.

Finally, we note that we can validate the  $\Delta \Delta G_{ij,\ell}$  calculations by constructing the cycle for transformations between three models TIP4P, TIP3P, and SPC/E represented by  $i, j, k$ , which must satisfy

$$\Delta G_{ij,\ell} + \Delta G_{jk,\ell} + \Delta G_{ki,\ell} = 0. \quad (7)$$

In our calculations, the free energy change  $\Delta G_{ij,\ell}$  is estimated along the path linearly varying all interactions (oxygen charge  $q$ , Lennard-Jones parameters  $C_6$  and  $C_{12}$ ) and geometry (bonds, bond angles, and location of the virtual charge site) parameters between models  $i$  and  $j$ .

All force field parameters for the water models used are given in Table I. We define a  $\lambda$  parameter which varies linearly from zero to one. A force field parameter set  $P$  for a state intermediate to two end states  $i$  and  $j$  can be defined as  $P(j, i, \lambda) = P_i(1 - \lambda) + P_j(\lambda)$ . We must then define mappings  $T_{ji}$  between any two states with parameters  $P(j, i, \lambda)$  which we describe in the case of specific transformations between water models in the supplementary material.<sup>32</sup>

For all simulations, we use Ewald order of interpolation of 4, Ewald tolerance of  $10^{-4}$ , Fourier spacing of 0.09 nm, Coulomb and Lennard-Jones cutoffs of 0.9 nm with a switch starting at 0.85 nm for both interactions as well as a short range neighbor list cutoff (rlist) of 1.1 nm, and a Fourier spacing of 0.1 nm. A Nosé-Hoover<sup>33</sup> thermostat was used with a time constant of 5 ps. Pressure was kept constant at 1.01325 bars (1 atm) using Martyna-Tuckerman-Tobias-Klein<sup>34</sup> barostat with time constant of 5 ps and compressibility of  $4.5 \times 10^{-5} \text{ bar}^{-1}$ . The time step was 0.002 ps, and bonds constrained using the SETTLE algorithm.<sup>35</sup>

A cubic box  $2.5 \times 2.5 \times 2.5 \text{ nm}^3$  containing 521 water molecules was used for each of the three water transformations. We performed 10 ns of NPT simulations at each of  $K = 41$  different equally spaced physical states for each transformation. Each of the  $K - 2$  intermediate states has force field parameters linearly interpolated between the two end state water models as described above. We record

TABLE I. Force field parameters for the water models used in this study.

| Water models →<br>Parameters ↓ | Units                    | SPC/E                     | TIP3P                     | TIP4P                    |
|--------------------------------|--------------------------|---------------------------|---------------------------|--------------------------|
| $C_6$                          | kJ/nm <sup>6</sup> /mol  | $2.616907 \times 10^{-3}$ | $2.489480 \times 10^{-3}$ | $2.55224 \times 10^{-3}$ |
| $C_{12}$                       | kJ/nm <sup>12</sup> /mol | $2.633237 \times 10^{-6}$ | $2.435088 \times 10^{-6}$ | $2.5104 \times 10^{-6}$  |
| $q(O)$                         | e                        | -0.8476                   | -0.834                    | -1.040                   |
| $q(H)$                         | e                        | 0.4238                    | 0.4170                    | 0.5200                   |
| $\angle \text{HOH}$            | deg                      | 109.47                    | 104.52                    | 104.52                   |
| $r_{OH}$                       | nm                       | 0.1                       | 0.09572                   | 0.09572                  |
| $r_{OM}$                       | nm                       | 0.000                     | 0.000                     | 0.015                    |
| $r_{HH}$                       | nm                       | 0.16330                   | 0.15139                   | 0.15139                  |

coordinates every 1 ps to create a trajectory of 10 000 coordinate samples, which are sufficiently far apart to have uncorrelated energies. We then have  $K$  trajectories worth of samples,  $\mathbf{X}_{jn}$ 's,  $j = 1 \dots K$  from these simulations at all states. A Gromacs trajectory conversion template was used to build a tool in C for operating directly on binary trajectory files to map the geometries of the simulated states to the trajectory of  $K - 1$  other states according to the transformation matrix defined in the supplementary material. This creates a matrix of  $K \times K$  trajectories consisting of samples  $T_{ji}(\mathbf{X}_{jn})$ . We then evaluate the energies of these trajectories using `mdrun` in rerun mode at the mapped geometries  $U_i(T_{ji}(\mathbf{X}_{jn}))$  to generate the  $\mathbf{U}[K \times K \times N_k]$  matrix of potential energies required for multistate reweighting calculations. The transformation  $T_{ji}(\mathbf{X}_{jn})$  and the subsequent rerun for a state  $i$  takes less than 5 min on a single processor, which is negligibly small compared to computational cost to generate  $\mathbf{X}_{jn}$ . From this process, we obtain  $\Delta G_{ij, \ell} \pm \delta(\Delta G_{ij, \ell})$  for  $N$  molecules. The molar value for the Gibbs free energy of transformation is  $\Delta G_{ij, \ell}/N$ , which means that the uncertainty will be  $\delta(\Delta G_{ij, \ell})/N$ , i.e., dividing by  $N$  and not by  $N^{1/2}$ , since changes in the molar value of each molecule are not independent.

For all water models, samples for estimating hydration free energies using standard alchemical methods by decoupling a single water molecule are generated for 21 intermediate states to solvate a water molecule in corresponding bulk liquid. These simulations are run for 20 ns to produce free energy of transfer estimates with useful uncertainties. Solute-solvent Coulomb interactions are decoupled linearly and the Lennard-Jones interactions are decoupled using a soft core interaction potential.<sup>31</sup>

### C. Dipole transformation

We used water models to test the mapping algorithm for estimating free energy differences between dissimilar rigid molecular geometries. In that case, part of the efficiency gain in computing the differences in free energies of solvation came from the fact that we were simulating pure liquid properties, and could divide by the number of molecules. We now wish to test the MBAR plus configurational mapping techniques for the free energy change for single molecule free energy differences. We also investigate whether the approach would work for dissimilar but still flexible geometries. We, therefore, also use this approach to compute the free energy difference between dipoles with different equilibrium lengths, and thus different dipole moments. In this case, there is some possibility of observing a dipole of a given equilibrium bond length in a simulation of a different equilibrium bond length, but the frequency is likely to be low if the equilibrium bond lengths are too different on the scale of the bond length fluctuations.

In this test, we calculate the free energy change for a change in the equilibrium bond length of a dipole solvated in TIP3P water while keeping all nonbonded interactions and the force constant of the harmonic bond unchanged. To model a dipolar solute, we consider a flexible dimer of two Lennard-Jones spheres (OPLS-UA ethane parameters) connected by a

harmonic bond, where charges of +1 and -1 are localized on the Lennard-Jones sites. This model is the same as the dipole model considered our previously developed molecular benchmark set.<sup>31</sup>

Again, the results from the standard alchemical decoupling approach for the difference in solvation free energy are compared to the results obtained by using MBAR with mapping between configurational ensembles. The configurations of the dipole corresponding to an equilibrium bond length are mapped to other equilibrium bond lengths according to a center of mass preserving transformation described in the supplementary material fractional deviation from the equilibrium length preserved.

The equilibrium bond length of dipole used in the initial state is 0.144 nm and in the final state is 0.164 nm, with force constant fixed at 224 262.4 kJ/mol/nm<sup>2</sup>. The estimated standard deviation in the distribution of bond lengths for this force constant is 0.0033 nm, so the equilibrium bond length difference between end states is 61 standard deviations. We simulate with nine equally spaced equilibrium dipole bond lengths intermediate to 0.144 and 0.164 to enable ample configurational space overlap so that free energy estimation using standard MBAR without any configurational mapping is possible. Mapping and re-evaluation of energies are done using the trajectories of the individual state simulations as described in Secs. II and III B. Each state was simulated for 10 ns. Other simulation parameters are the same as for the water models. In the case of the dipole exercise the exact same simulations are performed for both standard and mapped calculations; only the postprocessing analysis is different.

## IV. RESULTS

### A. Water models

#### 1. Free energy differences

We start with results for water model transformations, as the harmonic oscillator results are exact and were presented in Sec. III A. In Table II the hydration free energy  $\Delta G_{i, hyd}$  is calculated using MBAR without mapping for a single water molecule, using the decoupling method, for all three water models. Uncertainties in free energies are around 0.05 kJ/mol, and uncertainties in  $H_{vap}$  are around 0.002 kJ/mol, in both cases much smaller than experimental error. This high precision partly due is due to the large amount of simulation data used, nearly 400 ns in total for each water molecule free energy, a nontrivial amount even for a small water box of about

TABLE II. Solvation free energies,  $\Delta G_{hyd}$ , and enthalpy of vaporization  $H_{vap}$ .  $H_{vap}$  estimated using MBAR without mapping (column 3) has half the error as  $H_{vap}$  using the standard method of energy averages at the endpoint alone (column 4), as it uses information from nearby intermediate states.

| Water models | $\Delta G_{hyd}$<br>(kJ/mol) | $H_{vap}$ (MBAR)<br>(kJ/mol) | $H_{vap}$ (endpoint)<br>(kJ/mol) |
|--------------|------------------------------|------------------------------|----------------------------------|
| TIP4P        | $-25.375 \pm 0.047$          | $43.7589 \pm 0.0021$         | $43.7515 \pm 0.0042$             |
| TIP3P        | $-25.354 \pm 0.043$          | $42.4540 \pm 0.0017$         | $42.4507 \pm 0.0035$             |
| SPC/E        | $-29.302 \pm 0.046$          | $49.1172 \pm 0.0018$         | $49.1132 \pm 0.0042$             |

TABLE III.  $\Delta G_{ij,g}$  (column 1) is calculated analytically and  $\Delta G_{ij,\ell}$  (column 2) is calculated using MBAR with mapping. The differences  $\Delta\Delta G_{hyd}$  (decoupling) are calculated using direct subtraction and error propagation using data from Table II.  $\Delta\Delta G_{hyd}(\text{mapping}) - \Delta\Delta G_{hyd}(\text{decoupling})$  are zero within two and three standard deviations according to the thermodynamic cycle in Eq. (6) for all transformations. Subscripts  $a$ ,  $b$ , and  $c$  refer to the transformations labeled  $a$ ,  $b$ , and  $c$  in the first three rows. The result of decoupling cycle in column 5 of the last row is constrained to be identically zero numerically because it consists of quantities  $(i-j) + (j-k) + (k-i)$ . The result of the mapping cycle in column 4 of the last row is not constrained to be identically zero numerically as it consists of three independent calculations using different simulations; instead, it is in such close agreement with the thermodynamic cycle in Eq. (7) because of the high statistical precision of the approach. All free energies are in kJ/mol.

| Transformation                         | $\Delta G_{ij,g}$ | $\Delta G_{ij,\ell}$ | $\Delta\Delta G_{hyd}$<br>(mapping) | $\Delta\Delta G_{hyd}$<br>(decoupling) | $\Delta\Delta G_{hyd}$ (mapping) –<br>$\Delta\Delta G_{hyd}$ (decoupling) |
|----------------------------------------|-------------------|----------------------|-------------------------------------|----------------------------------------|---------------------------------------------------------------------------|
| (a) TIP4P→TIP3P                        | 0                 | $-0.0410 \pm 0.0002$ | $-0.0410 \pm 0.0002$                | $0.021 \pm 0.063$                      | $0.062 \pm 0.063$                                                         |
| (b) TIP3P→SPC/E                        | $-0.4570$         | $-4.4575 \pm 0.0001$ | $-4.0005 \pm 0.0001$                | $-3.948 \pm 0.063$                     | $-0.073 \pm 0.063$                                                        |
| (c) SPC/E→TIP4P                        | $0.4570$          | $4.4984 \pm 0.0002$  | $4.0414 \pm 0.0002$                 | $3.927 \pm 0.066$                      | $0.094 \pm 0.066$                                                         |
| $\Delta G_a + \Delta G_b + \Delta G_c$ | 0                 |                      | $0.0001 \pm 0.0003$                 | 0                                      |                                                                           |

500 molecules. The heat of vaporization endpoint calculations use only the fully coupled simulation, thus only 20 ns of simulation. The uncertainty in the heat of vaporization calculation using MBAR, utilizing the same data from the free energy calculations, is about half that of the enthalpy of vaporization using simulations only of the coupled endpoint. This is because simulations with similar potentials will have many configurations with non-negligible Boltzmann weight in the ensemble we are calculating our statistics at. Because error scales as  $N_{samp}^{-1/2}$  with  $N_{samp}$  the number of samples, it means that we are effectively extracting  $3 \times$  extra samples from the 21 intermediates that can be used to compute expectation averages at the fully coupled endpoint.

In Table III, we directly compare the results for the  $\Delta\Delta G_{i,hyd}$  generated using the standard single molecule decoupling cycle in Table II to the mapping cycle presented in this study, contrasted in columns 4 and 5, respectively, via the thermodynamic cycle in Eq. (6). Clearly the mapping cycle calculations are far more efficient, with uncertainties at least two orders of magnitude lower. The differences between the two cycles for the calculation are shown in the last column, and are all well within two standard deviations. Since the statistical error is essentially entirely in the alchemical results, we need additional ways of validating the high predicted precision of these mapping based calculations.

We therefore examine the cyclic transformation between the three water models described in thermodynamic cycle in Eq. (7) for the mapped results in column 4 of the last row, and see that it is zero with extremely high statistical precision; within  $3 \times 10^{-4}$  kJ/mol, demonstrating the extremely low estimated statistical errors, all near  $10^{-4}$  kJ/mol, are in fact accurate. The result of alchemical cycle in column 5 of the last row is constrained to be identically zero numerically because it consists of quantities  $(i-j) + (j-k) + (k-i)$ . However, the result of the mapping cycle in column 4 of the last row is not constrained to be identically zero numerically, as it consists of numbers generated from three independent simulations; instead, it is in such close agreement with the thermodynamic cycle in Eq. (7) because of the efficiency of the method.

Figure 2 shows the free energy differences as a function of distance along the coupling term between states. Figure 3 shows convergence of free energy estimate using different number of intermediate states, with a fixed amount (10 ns) of sampling. In both cases, we see that the results converge to the correct value qualitatively with only one intermediate state; even with no intermediate states, the free energies are quite close to the limit with more states. More quantitatively, Figure 2 shows that the value is indistinguishable to the final results with only five intermediate states.

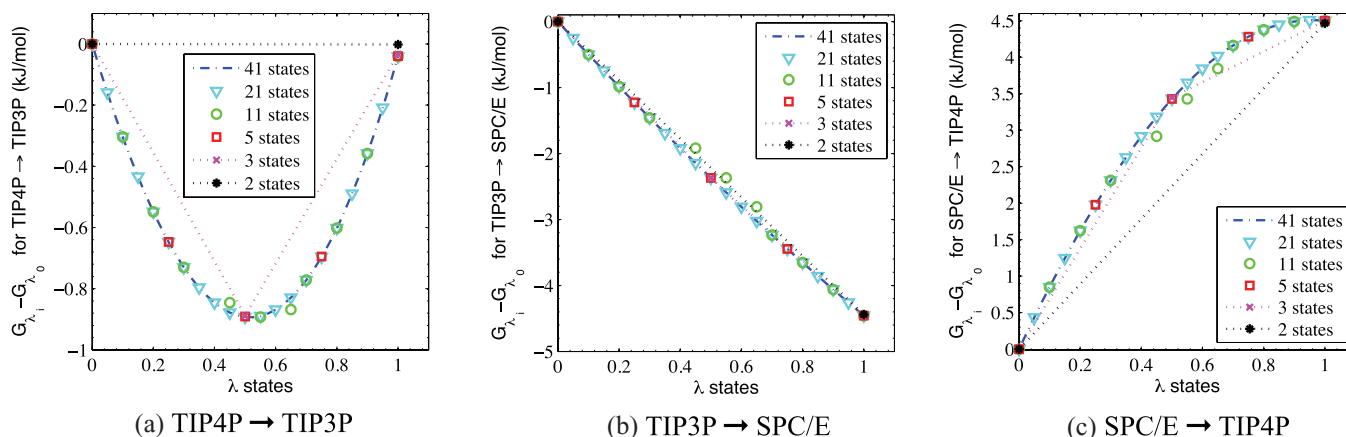


FIG. 2. Free energy differences along the transformation coordinate are plotted for the three water transformations. Three intermediate states are sufficient to accurately estimate the free energy difference, or a single intermediate state if less precision is required. Uncertainties are the same size as or smaller than the symbols.

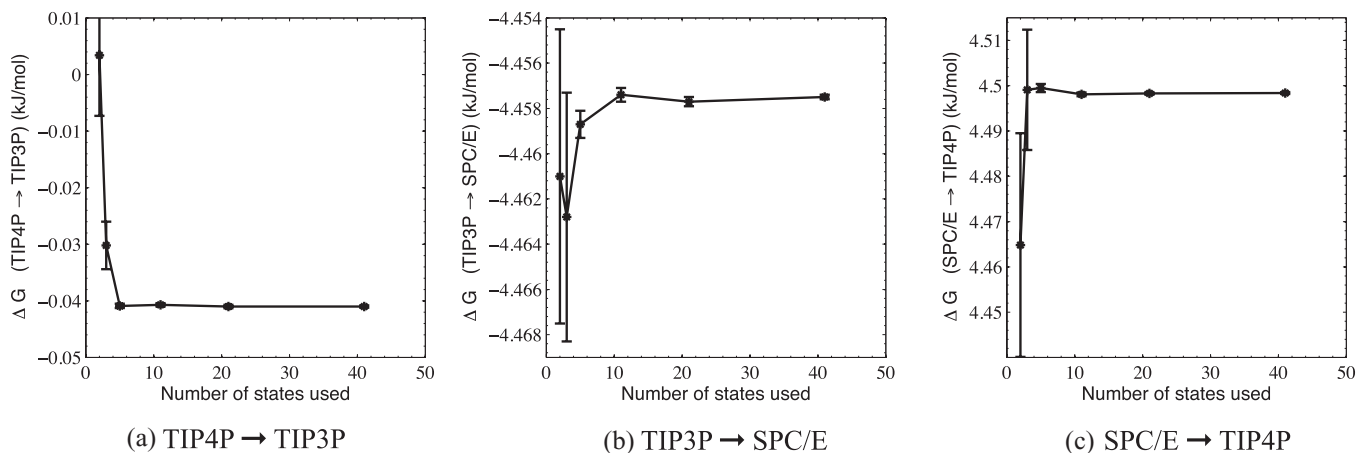


FIG. 3. Free energy estimates converge with just five intermediate states for all the transformations.

Clearly, in most cases there is no need to compute free energies to within  $10^{-4}$  kJ/mol. Instead, these more efficient methods can be used to calculate thermodynamics properties with far less sampling, either reducing the number of intermediates or reducing the length of simulations, drastically improving the throughput of calculations. In Figure 4(a), we compare both of the methods as a function of the number of samples used and the number of states along the path simulated for the SPC/E to TIP4P calculation.

We examine calculations with 2, 3, 5, 11, and 21 states and 10, 100, 1000, and 10000 samples per state to calculate  $\Delta\Delta G_{hyd}$  between SPC/E and TIP4P models using MBAR with mapping. For all combinations of number of states and number of samples per state, uncertainty in  $\Delta\Delta G_{hyd}$  using MBAR with mapping is 2–3 orders of magnitude lower compared to the lowest uncertainty when MBAR without mapping (magenta) is used as seen in Figure 4(a). This  $\Delta\Delta G_{hyd}$  (from

Table III, column 5, row 3) is the highest precision decoupling result, using all the available data. In Figure 4(b), we plot the log of the statistical uncertainty of the calculations. If we consider 0.01 kJ/mol as a useful target uncertainty amount, we can achieve this with only five states and 100 samples per state; if 0.05 kJ/mol is sufficient, the level achieved by standard decoupling methods, then we require only ten samples using either one or no intermediates (though the bias becomes approximately the same size as the uncertainty in the case of no intermediates).

We can quantify this improvement in efficiency by estimating the number of samples required of method A to achieve similar precision as method B, assuming that we are in the large sample regime so that the statistical uncertainty scales as  $N^{-1/2}$ . In this case, then we can express an efficiency of method A compared to method B in terms of the number of samples  $N_A$  required to reach the same precision as method B

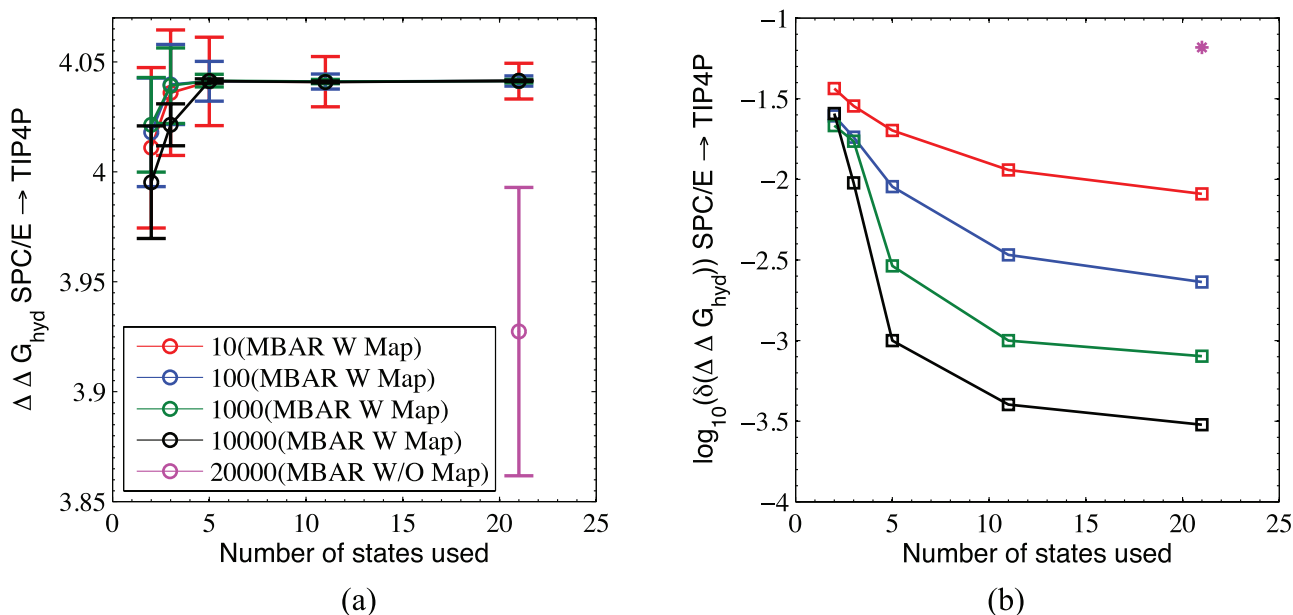


FIG. 4.  $\Delta\Delta G_{hyd}$  for SPC/E-TIP4P estimated using MBAR with mapping with different number of states (2, 3, 5, 11, 21) and number of samples per state (10, 100, 1000, 10000) has 2–3 orders lower uncertainty compared with  $\Delta\Delta G_{hyd}$  for SPC/E-TIP4P estimated using MBAR without mapping with 21 states and 20 000 samples per state.

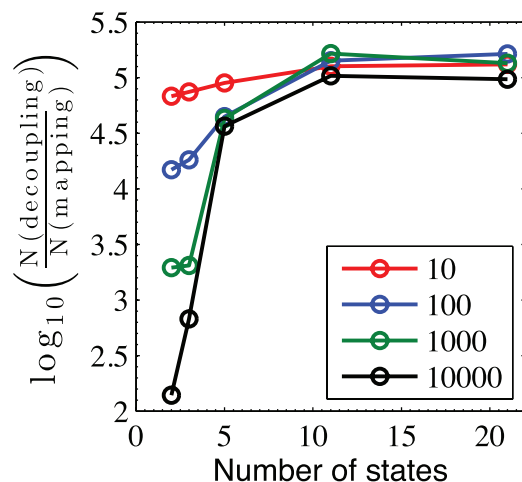


FIG. 5. Ratio of samples required for MBAR without remapping vs. with remapping to achieve a target statistical uncertainty. Mapping and reweighting approaches require  $10^2$ – $10^5$  times fewer samples compared to MBAR without mapping to achieve the same precision in  $\Delta\Delta G_{hyd}$ .

which used  $N_B$  samples, via the equation

$$\frac{N_A}{N_B} = \left( \frac{\delta(\Delta\Delta G_{hyd})_B}{\delta(\Delta\Delta G_{hyd})_A} \right)^2. \quad (8)$$

In Figure 5 we plot the log of the ratio,  $\log_{10}(N(\text{decoupling})/N(\text{mapping}))$ , as a function of number of states and number of samples per state. We clearly see that MBAR with mapping is statistically more efficient compared to the standard decoupling calculations without mapping by 2–5 orders of magnitude, and in most cases, closer to five orders of magnitude.

## 2. Enthalpy and entropy differences

Estimation of equilibrium observables using information from all intermediate states, of very different geometries, is also possible using MBAR with mapping. In this study, we specifically calculate the difference in enthalpy of vaporization and the difference in entropy of solvation for all three transformations. The difference in enthalpy of vaporization between two water models is equal to enthalpy of transformation,  $\Delta H_{ij,\ell}$ , since our transformations only involve wa-

ter at both end states. We neglect analytical correction factors that can be used to increase the accuracy of comparison to experiment,<sup>36</sup> as they do not affect the statistical uncertainty.

In Table IV column 2,  $\Delta H_{vap} = \Delta H_{ij,\ell}$  is evaluated using MBAR with mapping. In Table IV column 3,  $\Delta H_{vap}$  is calculated by subtracting the heats of vaporization in Table II evaluated by MBAR. Interestingly, unlike the case of  $\Delta\Delta G_{hyd}$ , we see that  $\Delta H_{vap}$  using the mapping and alchemical approaches agree within statistical error, with no improvement in the uncertainties for the mapping transformation. Part of the reason the efficiency of calculating  $\Delta\Delta G_{hyd}$  was so high is that the mapping techniques allowed us to use a transformation where we can calculate molar values, and divide the result by  $N$  molecules, which was not possible with standard decoupling techniques. However, for  $H_{vap}$ , we are already taking advantage of this efficiency gain.

There are other reasons that expectation values cannot be improved as much by increases in overlap as the free energies. As we see in the case of harmonic oscillators, improvements in the efficiency of expectation averages (such as the enthalpy) were due to the increase of effective numbers of samples at each state due to the non-negligible Boltzmann weighted contributions to the average from samples at nearby states. With the perfect mapping in the harmonic oscillator case, all samples contributed to expectation values at all thermodynamics states. This is not the case with water, however, as the overlap between intermediate states is much lower.

For endpoint reweighting methods, increasing the overlap between the two phases spaces by a factor of 10 decreases the variance roughly by the same factor of 10.<sup>6</sup> If the Boltzmann weights are initially 1%, then the effective contribution of samples from one state to the expectation values at neighboring states using reweighting is only 1% of the samples at the state itself. Increasing the overlap, and thus the Boltzmann weights of neighboring states 10% only increases the effective number of samples by 10%, for a total efficiency gain of  $1.10/1.01 = 1.09$  in expectations rather than a factor of 10 improvement for free energies. Expectation values cannot, therefore, be improved by as much as free energies can by increasing overlap between neighboring states. Because the exact quantification of overlap integrals in multistate simulations is an ongoing topic of research, it is not possible to make more quantitative comparisons, but the qualitative trend is clear.

TABLE IV. All enthalpies are in kJ/mol and all entropies are in J/mol/K. Decoupling  $\Delta H_{vap}$ s are calculated using direct subtraction and error propagation using data from Table II. The differences  $\Delta S_{ij,\ell}$  are calculated using the relationship  $\Delta S = (\Delta H - \Delta G)/T$  for both decoupling and mapping cases, with uncertainties propagated using the covariance calculated with MBAR. Subscripts  $a$ ,  $b$ , and  $c$  refer to the transformations labeled  $a$ ,  $b$ , and  $c$  in the first three rows. The thermodynamic cycle in Eq. (7) is satisfied for enthalpy  $H$  and entropy  $S$ . Enthalpies and entropies calculated with decoupling method agree with the ones calculated using mapping formalism within statistical error.

| Transformation                         | $\Delta H_{vap} = \Delta H_{ij,\ell}$<br>(mapping) | $\Delta H_{vap}$<br>(decoupling) | $\Delta S_{ij,\ell}$<br>(mapping) | $\Delta S_{ij,\ell}$<br>(decoupling) |
|----------------------------------------|----------------------------------------------------|----------------------------------|-----------------------------------|--------------------------------------|
| (a) TIP4P→TIP3P                        | $-1.3074 \pm 0.0034$                               | $-1.3049 \pm 0.0027$             | $-4.221 \pm 0.011$                | $-4.42 \pm 0.21$                     |
| (b) TIP3P→SPC/E                        | $6.6674 \pm 0.0028$                                | $6.6632 \pm 0.0025$              | $35.560 \pm 0.010$                | $35.37 \pm 0.21$                     |
| (c) SPC/E→TIP4P                        | $-5.3587 \pm 0.0031$                               | $-5.3583 \pm 0.0028$             | $-31.334 \pm 0.010$               | $-30.95 \pm 0.22$                    |
| $\Delta H_a + \Delta H_b + \Delta H_c$ | $0.0013 \pm 0.0054$                                | 0                                |                                   |                                      |
| $\Delta S_a + \Delta S_b + \Delta S_c$ |                                                    |                                  | $0.005 \pm 0.018$                 | 0                                    |

Additionally, the neighboring states contributing to the endpoint averages are somewhat different in the mapping case and in the decoupling case. In the decoupling case, intermediate states have  $N-1$  identical water molecules, and one water molecule being turned off; in the mapping case, intermediates have all of the molecules changing types slightly, with a mapping to increase phase space overlap. In the limit of fewer intermediates, which clearly is possible, then there will be reduced overlap between alchemical intermediate states. In the case of mapping, when the number of intermediates drops to five, then the  $\Delta H_{vap}$  results are statistically indistinguishable from using only the endpoints (see Table I in the supplementary material<sup>32</sup>). Fortunately, these endpoint results are already very accurate.

The entropy of transformation is calculated using the relationship  $\Delta S = (\Delta H - \Delta G)/T$ , with uncertainties in  $\Delta H$  and  $\Delta G$  propagated using MBAR. In Table IV columns 4 and 5, the entropy of transformation  $\Delta S_{ij, \ell}$  estimated using MBAR with mapping agrees within statistical error, with the entropy of transformation estimated using decoupling methods, for all three transformations. The uncertainty in entropy of transformation using mapping is an order of magnitude lower compared to the single molecule decoupling methods. We only use the MBAR enthalpy differences, because only when using MBAR are we able to accurately compute the correlations between the uncertainties in the free energies and the uncertainties in the enthalpies to propagate into uncertainties in the entropy. Clearly, these uncertainties are not independent, as they are calculated from the same data.

Although enthalpies have not changed significantly by adding mapping, the lower uncertainty of free energies estimated using mapping helps significantly in lowering the uncertainty in entropy of transformation. In standard procedures for calculating the entropy, obtaining low statistical error can be challenging because of the loss of statistical precision that occurs when subtracting free energies from enthalpies, which may be of similar magnitude, in order to obtain entropies directly. Similar loss of precision occurs when performing

finite difference calculations of free energies carried out at two close temperatures, as these free energies are certainly of the same magnitude. However, the case of our study, because of the extremely high precision of both enthalpy and entropy differences, we are able to extract entropy differences between water models with very low uncertainties, only 0.01 J/mol/K, with relatively little computational effort. The thermodynamic cycle in Eq. (7) is not only applicable for free energy of transformation but also for other state variables like enthalpy and entropy of transformation. The last two rows in Table IV demonstrate this fact by applying Eq. (7) for enthalpy and entropy of transformation and simultaneously validate these results.

We note that the hydration free energies using the decoupling methods do not involve transformation of all the molecules, and the uncertainties are, therefore, not scaled by the number of molecules. Part of the increase in efficiency using mapping is allowing us to use information from all of the molecules transforming, i.e., changing which cycle we calculate, and not solely using mapping to improve the convergence of a given cycle. Decoupling all the molecules simultaneously to the ideal gas state would have resulted in crossing a phase boundary during the transformation, which would lead to essentially zero overlap and poor efficiency. So although only part of the efficiency gain results *directly* from the mapping, the mapping between states also makes it possible to perform calculations not otherwise possible. One alternate method that could also take advantage of all-molecule transformation would have been a thermodynamic integration of all  $N$  molecules between states. However, as thermodynamic integration is not generally as efficient as MBAR,<sup>31</sup> and also requires an additional calculation of the change of holonomic constraints, such an approach was not pursued in this study.

### 3. Dipole transformation

We next examine the results of the free energy of dipole length change. Figure 6 and Table V show free energies

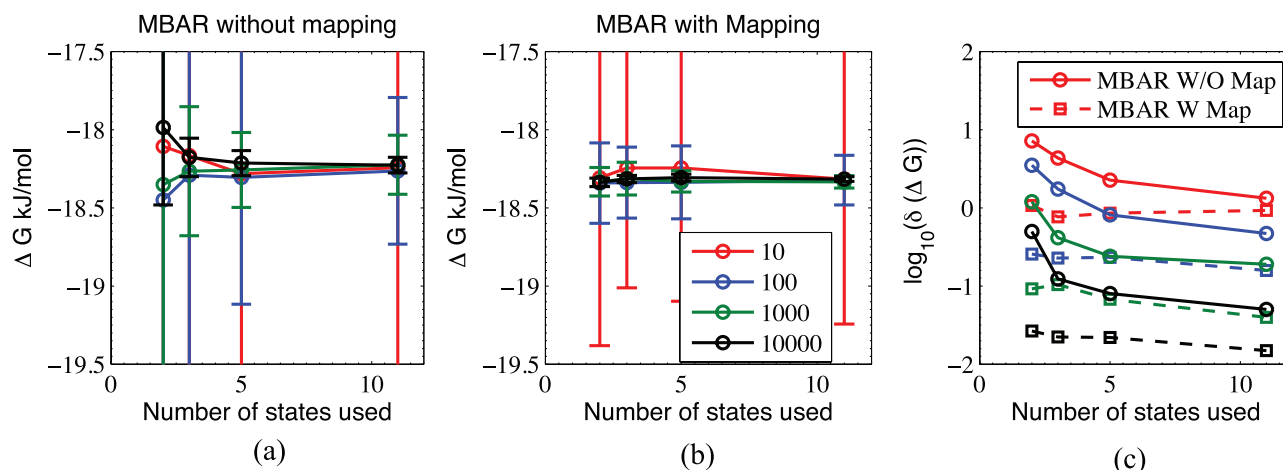


FIG. 6. The free energy differences between dipoles of different equilibrium lengths are calculated using two approaches: (a) MBAR without mapping and (b) MBAR with mapping. Different number of states (2, 3, 5, 11) and number of samples in each state (10, 100, 1000, 10 000) are used to estimate the free energies. MBAR with mapping estimates converged free energies with low uncertainty using only 2 states and 100 to 1000 samples per state whereas MBAR without mapping gives unconverged free energy estimate with high uncertainty unless many intermediates are used. In subplot (c) we see that the uncertainty in free energy using MBAR with mapping is an order lower compared to uncertainty in free energy estimated using MBAR without mapping using the same amount of sampling.

TABLE V. MBAR with mapping requires just 2 states and 1000 samples per state to estimate converged and precise free energy difference between dipoles of very different equilibrium lengths.

| States →  | 2                   |                     | 11                  |                     |
|-----------|---------------------|---------------------|---------------------|---------------------|
|           | (Mapping)           | (No mapping)        | (Mapping)           | (No mapping)        |
| Samples ↓ |                     |                     |                     |                     |
| 10 000    | $-18.337 \pm 0.027$ | $-17.986 \pm 0.496$ | $-18.316 \pm 0.015$ | $-18.227 \pm 0.050$ |
| 1000      | $-18.333 \pm 0.092$ | $-18.351 \pm 1.197$ | $-18.334 \pm 0.040$ | $-18.224 \pm 0.189$ |
| 100       | $-18.342 \pm 0.257$ | $-18.450 \pm 3.512$ | $-18.323 \pm 0.158$ | $-18.264 \pm 0.469$ |
| 10        | $-18.308 \pm 1.074$ | $-18.107 \pm 7.158$ | $-18.317 \pm 0.927$ | $-18.244 \pm 1.323$ |

between dipoles of different equilibrium bond lengths calculated using MBAR without mapping (Fig. 6(a)) and MBAR with mapping (Fig. 6(b)). In Fig. 6(c) we see that the MBAR with mapping shows significant improvement in the accuracy as well as precision in the free energy estimates over standard approach for same number of states and number of samples. The free energy differences are calculated using different number of states (2, 3, 5, 11) and different numbers of samples per state (10, 100, 1000, 10 000).

We next compare the statistical efficiency of MBAR with mapping against MBAR without mapping for each combination of number of states and samples per state in Figure 7, using Eq. (8). We plot the log of ratio of samples required for a given precision  $\log_{10}(N(\text{no mapping})/N(\text{mapping}))$  as a function of number of states and number of samples per state. Depending on the precise choice of intermediates states, for this calculation MBAR with mapping will require 3–300 times fewer samples compared to MBAR without mapping for the same precision in  $\Delta G$ . In this case, we see statistical efficiency improvements between 0.5 and 2.5 orders of magnitude, compared to 2–5 orders of magnitude in the case of hydration free energies. In this case, the gain in efficiency is solely because of mapping; the simulations are used for both calculations. For a 0.1 kJ/mol level of uncertainty, we see that 1000 samples with only the two end states are sufficient using mapping; it would require more than 250 000 samples using

standard techniques. For 0.25 kJ/mol, or about 0.06 kcal/mol uncertainty, which is frequently the level of experimental precision for accurate solvation free energies, only 100 samples are required with just the two endpoint states.

## V. DISCUSSION AND CONCLUSIONS

In this study, we have extended the capacity of multi-state reweighting to estimate thermodynamic observables for alchemical transformations involving changes in geometry. The multistate Bennett acceptance ratio method (MBAR) is coupled with configuration mapping to derive the new approach. The approach has been validated and tested using a toy model of truncated harmonic oscillators, the interconversion of SPC/E, TIP3P, and TIP4P water models, and the free energy difference of change in equilibrium bond length of a dipole. Mapping between mutually inaccessible configurations and speedup achieved by using reweighting combined with mapping can enable very high-throughput thermodynamic property prediction required for exploring large parameter and chemical spaces.

The mapping algorithm estimates the free energy between the truncated harmonic oscillators with zero error, independent of the number of samples. Estimates of free energy, enthalpy, and entropy of transformation of water molecules are verified by the thermodynamic cycles within two to three standard deviations. Estimates of free energy of transformation of dipole equilibrium bond length also agree with results obtained from alchemical approach within two to three standard deviations. In both the water model and dipole examples, we require very few samples and intermediate states if we use mapping with multistate reweighting. The mapping algorithm improves the statistical efficiency of thermodynamic property estimation by up to five orders of magnitude compared to standard decoupling approaches, with the advantage most pronounced for free energies and entropies.

The mapping technique can potentially be applied to force field parameterization or to any problem which involves estimation of thermodynamic observables over two or more states having poor or no overlap in configurational space, as long as an appropriate mapping can be constructed. For example, one could compute the free energy or entropy differences between a coarse grained potential with the same number of degrees of freedom, such as a rigid phenyl ring and a three or four site model with the same symmetry. One could compute the free energies and enthalpies of transformation between serine and cysteine, or between fluoro-, chloro-, and bromo-halogenates with very little error, using mapping to

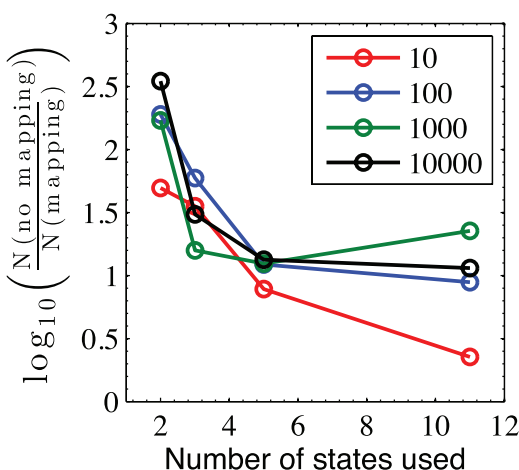


FIG. 7. Log ratio of samples required for MBAR without remapping vs. with remapping to achieve a target statistical uncertainty for the free energy difference between dipoles of different lengths. Phase space remapping combined with MBAR requires 3–300 times fewer samples compared to MBAR without mapping to achieve the same precision in estimating free energies between dipoles of different length.

account for bond length differences. It would be less useful for transformations where the changes in the energy terms are large enough to dominate the weightings of individual states even when the geometries are similar. For example, the energy differences resulting from a change from dipole to monopole electrostatics would likely dominate the computation of free energy differences when “morphing” from a water molecule to an ion.

Deriving an appropriate invertible mapping is a limiting factor but simplicity in simulation implementation, high efficiency of information usage and tremendous reduction in number of samples required for analysis should encourage future applications of this approach. Generally, molecular mappings that maintain the center of mass will be more straightforward, as the center of mass and intramolecular partition functions will be separable. Mappings that minimize changes in excluded volume will also increase the overlap, so shifts of atoms by only 10%–20% of the Lennard-Jones radii will be more efficient than mappings that drastically change excluded volume potentials. Aligning atomic sites in the end states to the extent possible is therefore recommended. In the supplementary material,<sup>32</sup> we outline in detail the transformation process used to map between water models as an example.

We also emphasize that this approach works well when the end state configuration ensembles differ explicitly in the geometric parameters used to define the states, such as bond lengths and angles. In the case of representing a complex many-body system evolving on a multidimensional landscape in a mean field way, the identification of a proper coordinate transformation, if any, is generally an intractable problem. Most mean field approaches use an approximation with a simplified energy function, but not necessarily different geometric configurations. Standard energy reweighting methods would be the most appropriate in those cases.

We emphasize that this analysis technique does not alter the collection of data at all, only the analysis. The time required to sample each new uncorrelated samples when performing simulations remains unchanged. If uncorrelated samples cannot be obtained at all, this method will still have bias. The reduction of error is because the number of correlated samples required to converge the physical quantity of interest is greatly decreased.

A number of avenues are possible to explore further using this formalism. Improving the estimation of observables between phase spaces with different discrete symmetries or that have changing numbers of degrees of freedom are beyond the scope of this paper, but would be very interesting to address with this formalism. The extension to differences in discrete symmetries would be likely be straightforward, through the inclusion of proper symmetry factors, though mapping between different degrees of freedom would not be as simple.

In this study, we have also restricted ourselves to configuration invariant transformations that preserve the center of mass. However, we could use transformations that are coordinate dependent, such as transformations that map between dihedral angle distributions of different amino acid side chains. The only difference is that since these transformations are coordinate dependent, the term must be directly added before performing multistate reweighting, instead of being added on

after the analysis is performed. Another possibility would be transformations of the locations of the molecular centers of mass, which would make it possible to construct maps between molecular systems of different average densities, or between NVT simulations performed at different volumes. Additional transformations beyond those discussed here are likely to yield even more interesting and effective approaches to estimating differences in thermodynamic properties as a function of molecular transformation.

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## APPENDIX: DERIVATION OF MULTISTATE MAPPING FORMALISM

The ratio  $r$  of normalization constants  $c_i$  and  $c_j$  between any two unnormalized probability distributions  $q_i$  and  $q_j$  defined on the same phase space can be related to the identity Eq. (A1)

$$r = \frac{c_i}{c_j} = \frac{E_j[q_i \alpha_{ij}]}{E_i[q_j \alpha_{ij}]}, \quad (\text{A1})$$

where  $E_i$  is the statistical expectation value of a system sampled using  $q_i$  and  $\alpha_{ij}$  is any strictly positive function defined on that same phase space, as derived in previous work on multistate reweighting techniques.<sup>4</sup>

For physical problems of thermodynamic interest, then  $q_i = \exp(-u_i(x))$  will be the Boltzmann weight, using the same definition of the reduced potential  $u_i(x)$  as in Ref. 4 to incorporate external parameters such as temperature, pressure, and chemical potentials. We can relate this ratio directly to the free energy and partition functions of the system:

$$\Delta A_{ij} = -k_B T \ln r = -k_B T \frac{Q_i}{Q_j}. \quad (\text{A2})$$

Although we use the canonical ensemble here, corresponding relationships hold for other ensembles. If we substitute the empirical estimator  $N_i^{-1} \sum_{n=1}^{N_j} g(x_{in})$  for the expectations  $E_i[g(x_{in})]$  in (A1), we can calculate these free energy differences from simulations,

$$r = \frac{\frac{1}{N_j} \sum_{n=1}^{N_j} q_i(x_{jn}) \alpha_{ij}(x_{jn})}{\frac{1}{N_i} \sum_{n=1}^{N_i} q_j(x_{in}) \alpha_{ij}(x_{in})}. \quad (\text{A3})$$

We now define a map  $T_{ij}$  which maps a configuration  $x_i$  sampled from the phase space  $\Gamma_i$  of state  $i$  to  $T_{ij}(x_i)$ , which is a sample contained in the phase space  $\Gamma_j$  of state  $j$  (again, the convention is that  $T_{ij}$  carries from phase space  $i$  to phase space  $j$ ). Assume that this map is completely invertible, such that  $T_{ij}^{-1} = T_{ji}$  exists and is one-to-one between phase space  $\Gamma_i$

and  $\Gamma_j$ .  $T_{ij}$  may depend on  $x_i$  (and likewise  $T_{ji}$  on  $x_j$ ) though our examples will not use this fact. The transformation will have an associated Jacobian factor  $J_{ij}$ , which also may be a function of  $x_i$ , and because the transformation is invertible, we will have  $J_{ij}(x_i) = J_{ji}^{-1}(x_j)$ .

We can sample from the unnormalized probability distribution function for  $j$  by performing sampling from state  $i$ , since

$$q_j = e^{-u_j(x_j)} = e^{-u(T_{ij}(x_i))} \quad (\text{A4})$$

and the partition function will be

$$c_j = \int_{\Gamma_j} e^{-u(x_j)} dx_j = \int_{\Gamma_i} e^{-u(T_{ij}(x_i))} |J_{ij}(x_i)| dx_i. \quad (\text{A5})$$

Based on similar arguments, Meng and Schilling<sup>30</sup> derived the equation for ratio  $r$  under the transformation for two states in the general statistical context. The equation from their paper can be rewritten for any pair of states  $i$  and  $j$  as

$$\frac{c_i}{c_j} = \frac{\frac{1}{N_j} \sum_{n=1}^{N_j} q_i(T_{ji}(x_{jn})) |J_{ji}(x_{jn})| \alpha_{ij}(x_{jn})}{\frac{1}{N_i} \sum_{n=1}^{N_i} q_j(T_{ij}(x_{in})) |J_{ij}(x_{in})| \alpha_{ij}(x_{in})}. \quad (\text{A6})$$

Here, the  $n$ th sample from the  $j$  state  $x_{jn}$  is mapped into the  $i$ th phase space using the map  $T_{ji}$ , and visa versa. Cross multiplying and summing both sides for  $j = 1$  to  $K$  in Eq. (A6),

$$\begin{aligned} \sum_{j=1}^K \frac{c_i}{N_j} \sum_{n=1}^{N_j} q_j(T_{ij}(x_{in})) |J_{ij}(x_{in})| \alpha_{ij}(x_{in}) \\ = \sum_{j=1}^K \frac{c_j}{N_j} \sum_{n=1}^{N_j} q_i(T_{ji}(x_{jn})) |J_{ji}(x_{jn})| \alpha_{ij}(x_{jn}). \end{aligned} \quad (\text{A7})$$

Without the variable transformation, then the choice of  $\alpha_{ij}$  that minimizes the variance if this estimate has the following form,<sup>4</sup>

$$\alpha_{ij}(x) = \frac{N_j c_j^{-1}}{\sum_{k=1}^K N_k c_k^{-1} q_k(x)}. \quad (\text{A8})$$

Using the definition of the transformed probability function, we can re-write  $\alpha_{ij}$  using the transformations as

$$\alpha_{ij}(x_i) = \frac{N_j c_j^{-1}}{\sum_{k=1}^K N_k c_k^{-1} q_k(T_{ki}(x_i)) |J_{ki}(x_i)|}. \quad (\text{A9})$$

Following,<sup>4</sup> we can take  $c_i/N_i$  out of the summation and then switch the order of summation on the left-hand side of Eq. (A7), and then use our formula for  $\alpha_{ij}$  to include the

transformations,

$$\begin{aligned} \frac{c_i}{N_i} \sum_{n=1}^{N_i} \sum_{j=1}^K \frac{q_j(T_{ij}(x_{in})) |J_{ij}(x_{in})| N_j c_j^{-1}}{\sum_{k=1}^K N_k c_k^{-1} q_k(T_{ik}(x_{in})) |J_{ik}(x_{in})|} \\ = \sum_{j=1}^K \frac{c_j}{N_j} \sum_{n=1}^{N_j} \frac{q_i(T_{ji}(x_{jn})) |J_{ji}(x_{jn})| N_j c_j^{-1}}{\sum_{k=1}^K N_k c_k^{-1} q_k(T_{jk}(x_{jn})) |J_{jk}(x_{jn})|}. \end{aligned} \quad (\text{A10})$$

Equation (A10) can now be written as

$$\begin{aligned} \frac{c_i}{N_i} \sum_{n=1}^{N_i} \frac{\sum_{j=1}^K N_j c_j^{-1} q_j(T_{ij}(x_{in})) |J_{ij}(x_{in})|}{\sum_{k=1}^K N_k c_k^{-1} q_k(T_{ik}(x_{in})) |J_{ik}(x_{in})|} \\ = \sum_{j=1}^K c_j N_j^{-1} \sum_{n=1}^{N_j} \frac{N_j c_j^{-1} q_i(T_{ji}(x_{jn})) |J_{ji}(x_{jn})|}{\sum_{k=1}^K N_k c_k^{-1} q_k(T_{jk}(x_{jn})) |J_{jk}(x_{jn})|}, \end{aligned} \quad (\text{A11})$$

which reduces to

$$c_i = \sum_{j=1}^K \sum_{n=1}^{N_j} \frac{q_i(T_{ji}(x_{jn})) |J_{ji}(x_{jn})|}{\sum_{k=1}^K N_k c_k^{-1} q_k(T_{jk}(x_{jn})) |J_{jk}(x_{jn})|}. \quad (\text{A12})$$

Taking negative log on both sides:

$$\begin{aligned} f_i &= -\ln(c_i) \\ &= -\ln \left( \sum_{j=1}^K \sum_{n=1}^{N_j} \frac{q_i(T_{ji}(x_{jn})) |J_{ji}(x_{jn})|}{\sum_{k=1}^K N_k c_k^{-1} q_k(T_{jk}(x_{jn})) |J_{jk}(x_{jn})|} \right), \end{aligned} \quad (\text{A13})$$

where  $f_i$  is the dimensionless free energy. Assuming that  $q(x) = \exp(-u(x))$  is the Boltzmann weight, we have

$$f_i = -\ln \left( \sum_{j=1}^K \sum_{n=1}^{N_j} \frac{e^{-u_i(T_{ji}(x_{jn}))} |J_{ji}(x_{jn})|}{\sum_{k=1}^K N_k e^{f_k - u_k(T_{kj}(x_{jn}))} |J_{kj}(x_{jn})|} \right). \quad (\text{A14})$$

We can define a “warped potential energy” term which includes the Jacobian term:

$$u_{ij}^w(x_{jn}) = u_i(T_{ji}(x_{jn})) - \ln(|J_{ji}(x_{jn})|). \quad (\text{A15})$$

Note specifically that the reduced potential includes a factor of  $\beta_i$ , while the Jacobian term does not. Note also that we must transform the samples from the  $j$ th thermodynamic state before evaluating the potential function in state  $i$ . We can then re-write the multistate reweighting equation in terms of

warped potential energy terms:

$$f_i = -\ln \left( \sum_{j=1}^K \sum_{n=1}^{N_j} \frac{\exp(-u_{ij}^w(x_{jn}))}{\sum_{k=1}^K N_k \exp(f_k - u_{kj}^w(x_{jn}))} \right). \quad (\text{A16})$$

Regaining the same form for the multistate Bennett acceptance ratio. Once we define the mappings and have  $K \times K \times N_k$  warped reduced energies, we can solve Eq. (2) self-consistently to find all free energies using methods already introduced.<sup>4</sup>

We next turn to expectation values of other observables computing using reweighting. This translation is trivial, because all other results will follow directly the equations above. We have previously shown<sup>4</sup> that expectations can be treated as probability distributions with no samples, and that if there are no samples collected, the math does not require that these quasi-probability distributions be non-negative. The only caution that must be followed is that observables must be computed using the transformed coordinates,  $A(T_{ij}(x_i))$ , which may require non-negligible extra computation for each sampled configuration.

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